

DRIVER DROWSINESS DETECTION

Lavi Garg
Dept. of Information Technology
Dharmsinh Desai University
Nadiad, Gujarat, India
21tuon138@ddu.ac.in

Charmi Parmar
Dept. of Information Technology
Dharmsinh Desai University
Nadiad, Gujarat, India
21itubs073@ddu.ac.in

Prof. Archana N. Vyas
Dept. of Information Technology
Dharmsinh Desai University
Nadiad, Gujarat, India
anvyas.it@ddu.ac.in

Abstract—Driver drowsiness is a major contributor to traffic accidents, highlighting the need for effective detection systems to improve road safety. This research presents a Driver Drowsiness Detection model based on a custom Convolutional Neural Network (CNN) architecture, designed to identify drowsiness from static facial images. The dataset used for training and evaluation was a combination of self-collected images from real-world conditions, including bus drivers, and publicly available datasets to ensure diversity. The custom CNN model was developed to achieve high accuracy in detecting drowsiness, reaching an accuracy of 81.73%. The implementation of this model can help reduce the risk of accidents and improve overall road safety.

Keywords—Convolutional Neural Network (CNN), Road Safety, Deep Learning, Image Classification, Accident Prevention, Custom CNN Model

I. INTRODUCTION

Driver drowsiness is a significant contributor to road accidents worldwide, making its early detection crucial for enhancing road safety. When drivers experience drowsiness, their attention and reaction times are severely impaired, increasing the likelihood of accidents. Traditional methods of detecting driver drowsiness, such as manual observation or physiological sensors, are often limited by human error, invasiveness, and the difficulty of monitoring drivers in real-time. To overcome these challenges, this research focuses on leveraging deep learning and computer vision technologies to develop a non-invasive and accurate system for detecting driver drowsiness through facial images. This approach offers a practical and scalable solution to address a widespread issue, potentially reducing accidents and saving lives.

The proposed system employs Convolutional Neural Networks (CNNs) to analyze facial features and detect signs of drowsiness. CNNs are highly effective for image classification tasks due to their ability to automatically extract relevant features from raw image data. In this research, facial cues such as eye closure, yawning, and head positioning are used to distinguish between drowsy and awake drivers. The system's ability to work with static images makes it an efficient solution for continuous monitoring without the need for specialized sensors or active driver involvement. This allows for seamless integration into existing in-vehicle systems, ensuring minimal disruption to the driver while maintaining reliable performance in real-time.

The system is trained using a custom dataset, which includes images collected from bus drivers under real-world conditions, such as varying lighting, facial occlusions, and demographic diversity. Additionally, publicly available datasets are incorporated to enhance the model's ability to generalize to different driving environments. The use of facial features enables the model to reliably classify drivers as either drowsy or awake, making it a suitable tool for early detection and intervention in real-time situations. This comprehensive dataset helps the model adapt to various challenges

encountered in practical applications, ensuring robustness and accuracy across different driving scenarios.

The research also focuses on improving the robustness of the model to handle different environmental factors that can affect the accuracy of drowsiness detection. Factors such as lighting changes, face orientations, and the presence of facial accessories are considered to ensure that the system performs well across diverse conditions. By accounting for these challenges, the system is designed to be reliable even in complex, real-world driving environments, enhancing its potential for widespread use. This level of adaptability is crucial for ensuring the system's effectiveness in a variety of settings, making it a versatile solution for both commercial and private vehicles.

The primary goal of this research is to develop a system that is both accurate and non-intrusive, capable of detecting driver drowsiness with high reliability. The scope of the project involves data collection, model development, and thorough evaluation using performance metrics such as accuracy, precision, recall, and F1-score. By incorporating this system into vehicles, it has the potential to reduce the number of accidents caused by drowsiness, ultimately contributing to improved road safety and helping prevent drowsy driving-related incidents. With further development, this system could become an essential safety feature in modern vehicles, significantly enhancing driver safety.

II. LITERATURE REVIEW

First Driver drowsiness detection has been an essential research area due to its critical role in enhancing road safety. Various methods and systems have been proposed over the years to detect and prevent accidents caused by driver drowsiness. Among the most commonly used techniques are those based on facial expressions, eye movements, head poses, and other indicators.

Ayman Altameem et al. [1] used Support Vector Machines (SVM) to detect facial expression changes, achieving 83.25% accuracy under varying lighting conditions. This demonstrated the potential of facial expressions for real-time detection.

Yaman Albadawi et al. [2] combined Eye Aspect Ratio (EAR) and head pose estimation, achieving 99% accuracy with classifiers like random forest and SVM. This approach enhanced the reliability of real-time drowsiness detection.

Hashemi et al. [3] introduced a CNN-based system focusing on eye closure detection. Their model, tested on the ZJU dataset, achieved 98.15% accuracy and low computational complexity, making it suitable for real-time applications.

Rateb Jabbar et al. [4] developed a CNN-based driver drowsiness detection system using facial landmark detection. Their model achieved 83.3% accuracy, utilizing the NTHU Driver Drowsiness Detection dataset. Designed for mobile

and embedded systems, it demonstrated real-time feasibility and enhanced road safety.

Salman et al. [5] proposed an ensemble CNN (ECNN) model that incorporated facial features like yawning and eye closure. Their system achieved 99.42% accuracy on the YawDD dataset, proving that ensemble learning can improve robustness in drowsiness detection.

Md. Tanvir Ahammed Dipu et al. [6] used MobileNet with Single Shot Multibox Detector (SSD) for detecting yawning and eye states, achieving a mean average precision (mAP) of 98.02%. This lightweight model was ideal for real-time detection.

Minhas et al. [7] tested InceptionV3, VGG16, and ResNet50 on a custom dataset, with ResNet50 achieving the highest accuracy of 93.69%. Their research demonstrated how CNNs can be adapted for diverse environmental conditions.

Jahan et al. [8] created a driver drowsiness detection system using a custom 4D CNN model, along with VGG16 and VGG19. The 4D model achieved 97.53% accuracy, while the VGG models reached 95.93% and 95.03%, respectively.

Esra Kavalcı Yılmaz and M. Ali Akçayol [9] introduced a hybrid VGG19+LSTM model, achieving 90.53% accuracy on the SUST-DDD dataset. This approach underscored the role of hybrid models in enhancing drowsiness detection.

Florez et al. [10] utilized ResNet50V2 for drowsiness detection and achieved 99.71% accuracy on the NITYMED dataset. Their use of transfer learning and MediaPipe for eye detection improved real-time performance.

Mohammed Imran Basheer Ahmed et al. [11] focused on facial expression analysis using CNNs and VGG16. Their system, tested on 2,900 images, achieved 97% accuracy, demonstrating CNN's potential in real-time detection.

Salem and Waleed [12] used MobileNetV2 and InceptionV3 models to develop a highly effective system, achieving accuracies of 99.94% and 99.82% on the Driver Drowsiness Dataset, proving their real-time applicability.

III. PROPOSED WORK

A. Dataset

The dataset used for this project is a combination of a custom-created dataset and publicly available sources from Kaggle. By merging these sources, the dataset gained substantial diversity, making it more applicable to real-world scenarios. It includes images captured under different lighting conditions (day and night), with facial obstructions like sunglasses, specs, masks, and hats, and from various camera angles, simulating a typical driving environment. The data is evenly distributed between two classes—Awake and Drowsy, with 5,000 images in each category.

A total of 7,258 images were self-collected from 27 subjects (14 male, 13 female) aged between 20 and 60 years, with some images specifically of bus drivers (300 images) to simulate different vehicle types. The remaining 2,742 images were sourced externally to complement the custom data. The combination of self-collected and external data ensures a balanced and varied dataset, allowing the model to generalize effectively in diverse conditions. This diverse dataset aims to improve the model's ability to detect drowsiness accurately

across a wide range of real-world driving environments. use abbreviations in the title or heads unless they are unavoidable.

B. Model Analysis

In our driver drowsiness detection system, we experimented with both pre-trained models (ResNet50, InceptionV3, and VGG16) and a custom-designed Convolutional Neural Network (CNN) tailored specifically for this task. The pre-trained models were fine-tuned on our dataset, leveraging their extensive feature extraction capabilities, which provided a strong baseline. These models benefited from their ability to capture complex patterns due to their deep architectures, having been trained on vast image datasets like ImageNet. However, while the pre-trained models showed promising results in terms of accuracy, their performance was constrained by the fact that they were not initially designed for detecting driver drowsiness. The transfer learning approach provided a quick way to adapt these models to the task, but the generalized features they learned did not fully capture the specific nuances of the drowsiness detection problem.

However, the custom CNN, which was specifically built for this project, outperformed the pre-trained models in terms of accuracy and classification performance. This custom architecture was optimized to capture the unique facial features associated with drowsiness, such as eye closure patterns, yawning, and subtle head movements. By fine-tuning the architecture and hyperparameters, the custom CNN was able to learn task-specific features more effectively, leading to more accurate predictions of 'Awake' and 'Drowsy' states. Additionally, the custom model was able to adapt better to the diverse environmental conditions present in the dataset, such as varying lighting and facial orientations, which made it more robust in real-world scenarios.

C. System Architecture

The system architecture for the driver drowsiness detection system is designed as a streamlined pipeline that processes input images to determine the driver's alertness. It starts with the data input phase, where facial images of the driver are captured in real-time through an onboard camera. These images are then passed through a preprocessing stage, where they are resized, normalized, and adjusted for variations such as lighting conditions and facial orientations to ensure consistency. The preprocessed images are then fed into the model for inference, where a Convolutional Neural Network (CNN), either custom-built or pre-trained, analyzes the facial features—such as eye closure and head position—to detect signs of drowsiness. It comprises three key components: data input, preprocessing, and model inference, followed by the final output, where the system classifies the driver as either 'drowsy' or 'awake.'

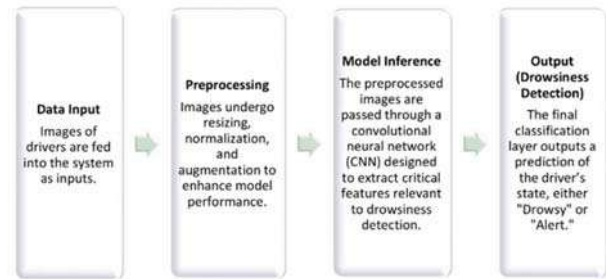


Figure 1 - System Architecture

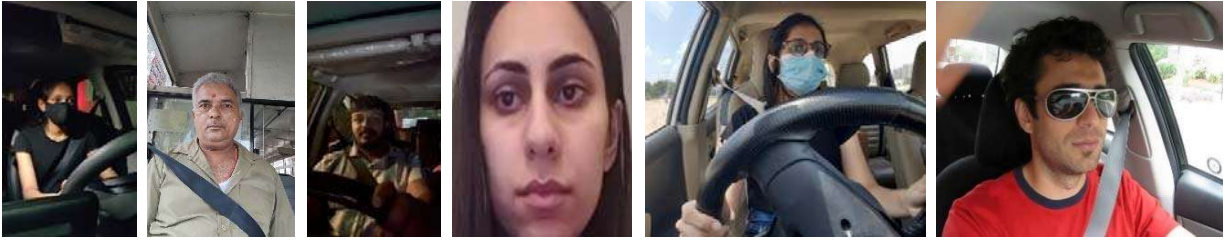


Figure 2 - Awake

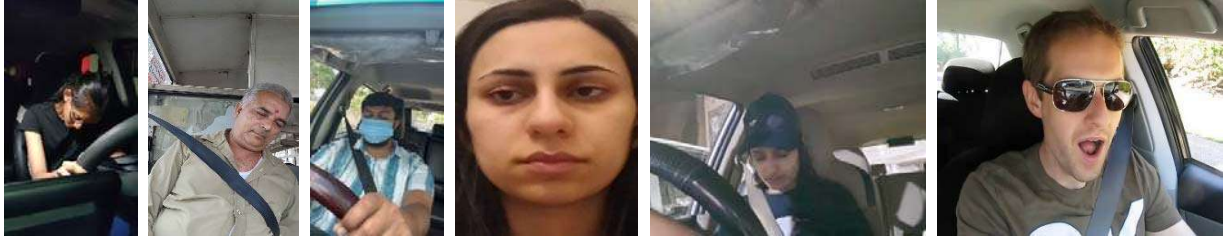


Figure 3 - Drowsy

D. Data Preprocessing and Model Training

a) Data Collection and Split

The dataset contains images of drivers in two states: Drowsy and Awake, and is divided into three sets: 70% for training, 15% for validation, and 15% for testing. The data is split using the `train_test_split` function from `scikit-learn`.

b) Image Preprocessing and Augmentation

All images are resized to 224x224 pixels and converted to RGB format for consistency. Face detection is performed using the Haar Cascade Classifier to focus on facial features, and if a face is detected, the image is cropped to highlight the face region; otherwise, the full image is used. To enhance the model's generalization capabilities, data augmentation techniques such as shear, zoom, horizontal flip, and brightness adjustments (ranging from 0.5 to 1.5) are applied to the training set, allowing the model to better handle variations in real-world scenarios.

E. Architecture Design

The CNN model for driver drowsiness detection follows a sequential architecture, designed to progressively extract and process features from input images. The model takes images of size (224, 224, 3) as input and applies a series of convolutional layers to learn and capture local features such as edges, textures, and shapes. These layers are followed by ReLU (Rectified Linear Unit) activation functions, which introduce non-linearity into the model, allowing it to model more complex relationships between the pixels in the image. To manage the size of the data and reduce computational load, max-pooling layers are employed after each convolution. These layers downsample the feature maps by selecting the maximum values in a given window, which also helps to minimize overfitting by preserving only the most important features.

As the data moves deeper into the model, the fully connected layers take the extracted features and map them to the final prediction. The model uses Softmax in the output layer to assign probabilities to each class—"Drowsy" or

"Awake." To further optimize training, the Adam optimizer is chosen for its adaptive learning rate capabilities and its efficiency in dealing with sparse gradients. Categorical cross-entropy is used as the loss function, comparing the predicted output probabilities with the true labels to measure the model's performance. This choice of loss function is well-suited for multi-class classification problems like this one.

To prevent overfitting, especially in the fully connected layers, dropout layers are included, randomly disabling a fraction of neurons during training to encourage generalization. The model was trained for 35 epochs, which provided a balance between achieving high accuracy and avoiding overfitting. Early stopping was also implemented, monitoring validation loss with a patience of 5 epochs. This mechanism ensures that the model halts training once performance starts to degrade, ensuring the best possible generalization to unseen data without excessive training.

Layer (type)	Output Shape	Param #
conv2d_4 (Conv2D)	(None, 222, 222, 32)	896
batch_normalization_4 (BatchNormalization)	(None, 222, 222, 32)	128
max_pooling2d_4 (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_5 (Conv2D)	(None, 109, 109, 64)	18,496
batch_normalization_5 (BatchNormalization)	(None, 109, 109, 64)	256
max_pooling2d_5 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_6 (Conv2D)	(None, 52, 52, 128)	73,856
batch_normalization_6 (BatchNormalization)	(None, 52, 52, 128)	512
max_pooling2d_6 (MaxPooling2D)	(None, 26, 26, 128)	0
conv2d_7 (Conv2D)	(None, 24, 24, 256)	295,168
batch_normalization_7 (BatchNormalization)	(None, 24, 24, 256)	1,024
max_pooling2d_7 (MaxPooling2D)	(None, 12, 12, 256)	0
flatten_1 (Flatten)	(None, 36,864)	0
dense_2 (Dense)	(None, 128)	4,718,720
dropout_1 (Dropout)	(None, 128)	0
dense_3 (Dense)	(None, 1)	129

Total params: 5,189,187 (19.49 MB)
 Trainable params: 5,188,225 (19.49 MB)
 Non-trainable params: 960 (3.75 KB)
 Optimizer params: 2 (12.00 B)

Figure 4 - CNN Architecture

IV. EXPERIMENTAL RESULTS

The experiments conducted to evaluate the performance of the driver drowsiness detection model were performed using the custom CNN model, as well as several pre-trained models, including ResNet50, InceptionV3, and VGG16. The evaluation metrics used for assessing model performance include precision, recall, F1-score, accuracy, and loss for the training, validation, and test datasets.

The model's performance evolved over 35 epochs, as shown in the following analysis of training and validation accuracy and loss. Training accuracy steadily increased, reaching 82.50% by the 35th epoch, while training loss decreased and stabilized at 0.8250, indicating effective learning without overfitting. Validation accuracy followed a similar trend, ending at 81.93%, with validation loss gradually decreasing to 0.3385, suggesting strong generalization to unseen data. By the 35th epoch, the model had achieved a good balance between accuracy and loss, making it suitable for deployment in detecting driver drowsiness.

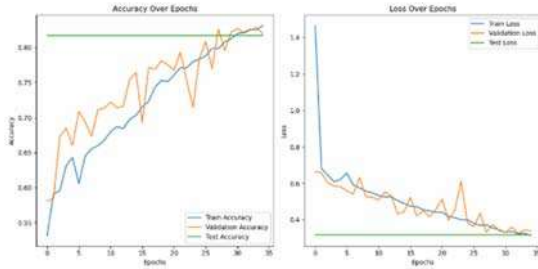


Figure 5 - Graph for accuracies and losses in CNN

The custom CNN model showed strong performance across training, validation, and test sets. During training, it effectively identified both "Awake" and "Drowsy" states with balanced precision and recall, suggesting good generalization to the training data. The model achieved solid accuracy and consistent F1-scores, indicating effective learning.

In the validation phase, the model maintained balanced precision and recall, excelling in detecting "Drowsy" states. The validation accuracy confirmed the model's ability to generalize well to unseen data, with slightly lower precision for "Drowsy," but overall consistent results.

On the test set, the model performed well, with high accuracy and a good balance between precision and recall. It showed greater sensitivity to identifying "Drowsy" states, although precision for "Drowsy" was lower than for "Awake." Overall, the model demonstrated robust generalization, making it suitable for real-world applications.

The confusion matrix for the test set reflects the model's ability to differentiate between "Awake" and "Drowsy" states. It shows the model's precision and recall performance for each class, indicating its ability to correctly identify both "Awake" and "Drowsy" drivers. The overall accuracy of the model on the test set is 81.73%, with a loss of 0.3172, highlighting its solid performance in real-world data classification.

	Awake (0)	Drowsy (1)
Awake (0)	554	185
Drowsy (1)	84	649

Table 1 - Confusion Matrix for CNN Test dataset

The ROC curve for the test dataset demonstrates the model's ability to distinguish between "Awake" and "Drowsy" states. The curve shows the trade-off between sensitivity (True Positive Rate) and specificity (1 - False Positive Rate). A higher AUC (Area Under the Curve) indicates better performance. For this model, the ROC curve reveals an AUC of 0.82, suggesting that the CNN model effectively distinguishes between the two states while maintaining a good balance between false positives and false negatives. This confirms the model's robustness and its ability to generalize well on unseen test data.

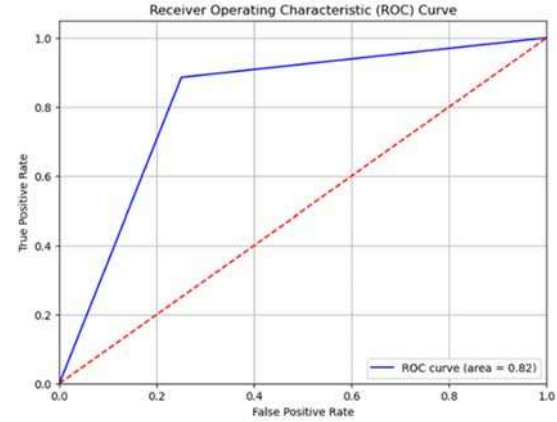


Figure 6 - ROC and AUC for CNN Test dataset

In addition to evaluating the custom CNN model, I also analyzed the performance of pre-trained models, including ResNet50, VGG16, and InceptionV3. These models were fine-tuned for the driver drowsiness detection task, and their performance was compared with the custom CNN. Each model's precision, recall, F1-score, and overall accuracy were recorded to understand their strengths and limitations.

Models	ResNet50	InceptionV3	VGG16	Custom CNN
Testing Accuracy	55.60%	77.00%	62.60%	81.73%
Validation Accuracy	52.60%	76.80%	64.00%	81.93%
Training Accuracy	51.24%	70.82%	59.01%	85%
Testing Loss	69.05%	48.11%	60.37%	31.72%
Validation Loss	69.24%	47.88%	59.80%	33.85%
Training Loss	69.25%	54.84%	64.60%	31.09%
Confusion Matrix	[[145, 355], [89, 411]]	[[393, 107], [123, 377]]	[[489, 11], [363, 137]]	[[554, 185], [84, 649]]
Precision (Awake)	0.62	0.76	0.57	0.87
Precision (Drowsy)	0.54	0.78	0.93	0.78
Recall (Awake)	0.29	0.79	0.98	0.75
Recall (Drowsy)	0.82	0.75	0.27	0.85
F1 Score (Awake)	0.40	0.77	0.72	0.80
F1 Score (Drowsy)	0.65	0.77	0.42	0.83

Table 2 - Final Comparison table for all models

The confusion matrices for all models highlight these performance differences. For example, ResNet50 and VGG16 had more misclassifications for "Drowsy" states, while InceptionV3 performed better but still fell short compared to the custom CNN. The custom CNN showed the most balanced confusion matrix, with fewer misclassifications in both "Awake" and "Drowsy" categories, making it the most reliable model for this task. These findings suggest that while pre-trained models provide a good starting point, the custom CNN achieves the best results.

To conclude, the analysis demonstrates that the custom CNN model outperforms others, achieving the highest testing accuracy of 81.73% with balanced precision and recall for both "Awake" and "Drowsy" classes. InceptionV3 also performs well, showing reliable accuracy and particularly excelling in detecting "Awake" states. VGG16 achieves good results for "Awake" detection but faces challenges with accurately identifying "Drowsy" states. ResNet50, however, shows lower accuracy and F1 scores, indicating limitations when applied to this dataset. Overall, the custom CNN model, tailored specifically for driver drowsiness detection, proves to be the most optimal, delivering reliable accuracy and robustness across various conditions.

V. CONCLUSION

This research project on Driver Drowsiness Detection successfully developed a custom CNN model that achieved an accuracy of 81.73% and an ROC AUC score of 0.82, showcasing its ability to identify drowsy drivers effectively. The custom CNN outperformed pre-trained models like ResNet50, VGG16, and InceptionV3, thanks to its tailored design. Key features such as convolutional layers with ReLU activation, max pooling, and dropout layers enabled the model to capture subtle facial cues related to drowsiness. The dataset, which included 72.59% self-collected data and 27.41% external images, ensured diversity and robustness. However, the model is limited to static image processing, which restricts its immediate use in real-time monitoring. Additionally, slight overfitting indicates the need for a more varied dataset to enhance generalization across different driving environments.

Future work will focus on expanding the dataset's diversity to improve generalization and reduce overfitting. The transition from static image processing to real-time video analysis is a key goal, enabling dynamic detection in active vehicle environments. Advanced techniques, such as transfer learning with video models and temporal data analysis, will be explored to improve accuracy and responsiveness. Additionally, optimizing the model's architecture for real-time deployment will ensure it can handle continuous data streams efficiently. By achieving these goals, the project has the potential to significantly enhance driver safety and prevent accidents caused by drowsiness. The next steps will ensure the model's practical application in real-world driving conditions, contributing to safer roads for all.

VI. REFERENCES

- [1] Altameem, Ayman, Ankit Kumar, Ramesh Chandra Poonia, Sandeep Kumar, and Abdul Khader Jilani Saudagar. "Early identification and detection of driver drowsiness by hybrid machine learning." *IEEE Access* 9 (2021): pp.162805-162819.
- [2] Albadawi, Yaman, Aneesa AlRedhaei, and Maen Takruri. "Real-time machine learning-based driver drowsiness detection using visual features." *Journal of imaging* 9, no. 5 (2023): 91.
- [3] Hashemi, Maryam, Alireza Mirrashid, and Aliasghar Beheshti Shirazi. "Driver safety development: Real-time driver drowsiness detection system based on convolutional neural network." *SN Computer Science* 1, no. 5 (2020): 289.
- [4] Pandey, Nageshwar Nath, and Naresh Babu Muppalaneni. "Real-time drowsiness identification based on eye state analysis." In 2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS), pp. 1182-1187. IEEE, 2021.
- [5] Salman, Rais Mohammad, Mahbubur Rashid, Rupal Roy, Md Manjurul Ahsan, and Zahed Siddique. "Driver drowsiness detection using ensemble convolutional neural networks on YawDD." *arXiv preprint arXiv:2112.10298* (2021).
- [6] Dipu, Md Tanvir Ahammed, Syeda Sumbul Hossain, Yeasir Arafat, and Fatama Binta Rafiq. "Real-time driver drowsiness detection using deep learning." *International Journal of Advanced Computer Science and Applications* 12, no. 7 (2021).
- [7] Minhas, Abid Ali, Sohail Jabbar, Muhammad Farhan, and Muhammad Najam ul Islam. "A smart analysis of driver fatigue and drowsiness detection using convolutional neural networks." *Multimedia Tools and Applications* 81, no. 19 (2022): 26969-26986.
- [8] Jahan, Israt, KM Aslam Uddin, Saydul Akbar Murad, M. Saef Ullah Miah, Tanvir Zaman Khan, Mehedi Masud, Sultan Aljahdali, and Anupam Kumar Bairagi. "4D: a real-time driver drowsiness detector using deep learning." *Electronics* 12, no. 1 (2023): 235.
- [9] Yilmaz, Esra Kavalci, and M. Ali Akcayol. "SUST-DDD: A real-drive dataset for driver drowsiness detection." In *Conference of Open Innovations Association (FRUCT)*, vol. 6. 2022.
- [10] Florez, Ruben, Facundo Palomino-Quispe, Roger Jesus Coaquira-Castillo, Julio Cesar Herrera-Levano, Thuanne Paixão, and Ana Beatriz Alvarez. "A cnn-based approach for driver drowsiness detection by real-time eye state identification." *Applied Sciences* 13, no. 13 (2023): 7849.
- [11] Ahmed, Mohammed Imran Basheer, Halah Alabdulkareem, Fatimah Alomair, Dana Aldossary, Manar Alahmari, Munira Alhumaidan, Shoog Alrassan, Atta Rahman, Mustafa Youldash, and Gohar Zaman. "A deep-learning approach to driver drowsiness detection." *Safety* 9, no. 3 (2023): 65.
- [12] Salem, Dina, and Mohamed Waleed. "Drowsiness detection in real-time via convolutional neural networks and transfer learning." *Journal of Engineering and Applied Science* 71, no. 1 (2024): 122.